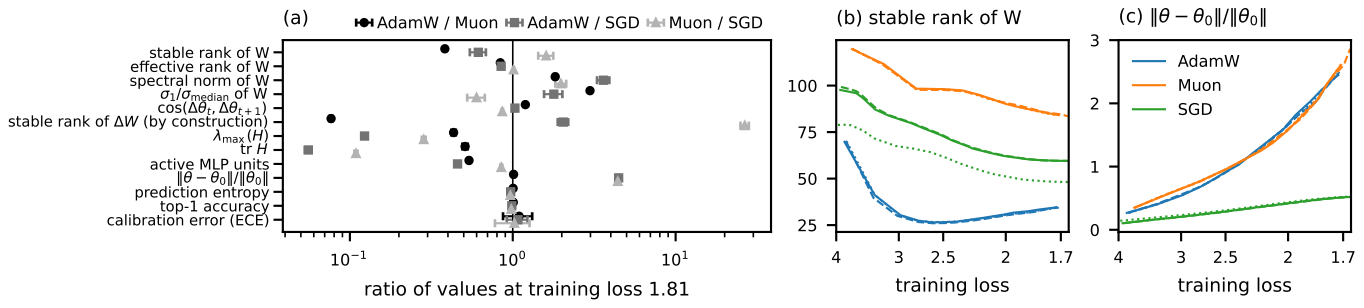
(a)
AdamW / Muon
AdamW / SGD
Muon / SGD
stable rank of W
effective rank of W
spectral norm of W
$\sigma_1/\sigma_{\text{median}}$ of W
$\cos(\Delta\theta_t, \Delta\theta_{t+1})$
stable rank of ΔW (by construction)
$\lambda_{\max}(H)$
tr H
active MLP units
$\|\theta-\theta_0\|/\|\theta_0\|$
prediction entropy
top-1 accuracy
calibration error (ECE)
10^{-1}
10^{0}
10^{1}
ratio of values at training loss 1.81
(b) stable rank of W
100
75
50
25
4
3
2.5
2
1.7
training loss
(c) $\|\theta-\theta_0\|/\|\theta_0\|$
AdamW
Muon
SGD
3
2
1
0
4
3
2.5
2
1.7
training loss